ASSIGNMENT – Question 2

Normal Distribution and Empirical Rule (Domain: Student Test Scores)

Introduction

In statistics, the **normal distribution** is one of the most important probability distributions used to represent real-world data. It describes how values are distributed around a central value called the mean. The graph of a normal distribution is symmetric and bell-shaped, which is why it is often called the **bell curve**.

The **empirical rule**, also known as the **68–95–99.7 rule**, explains how data is distributed within one, two, and three standard deviations from the mean. This rule helps in understanding how data spreads around the average value.

In this assignment, the selected domain is **student test scores**, as academic performance often follows a normal distribution pattern.

Domain Selection: Student Test Scores

The education domain is suitable for explaining normal distribution because:

- Most students score around the average.
- Some students score very high.
- Some students score very low.
- Extreme scores are rare.
- Performance varies due to preparation, understanding, and exam difficulty.

Because of these characteristics, student marks typically form a symmetric bell-shaped curve centered at the average score.

Understanding Normal Distribution

A normal distribution is completely defined by two parameters:

Mean (μ)

The mean represents the average score of all students.
If 100 students take a test and their average score is 70, then 70 is the mean.

Standard Deviation (σ)

Standard deviation measures how spread out the scores are from the mean.

- Small standard deviation → Scores are close to the average.
- Large standard deviation → Scores are widely spread.

Characteristics of Normal Distribution

- Bell-shaped curve
- Perfectly symmetric
- Mean = Median = Mode
- Highest frequency at the center
- Probability decreases as we move away from the mean

Empirical Rule (68–95–99.7 Rule)

The empirical rule explains how much data lies within different standard deviation ranges.

1. 68% Rule ($\mu \pm 1\sigma$)

Approximately **68% of students** score within one standard deviation from the mean.

Example:
Mean (μ) = 70
Standard Deviation (σ) = 10

Range: 60 to 80

This means most students score between 60 and 80.

2. 95% Rule ($\mu \pm 2\sigma$)

About **95% of students** score within two standard deviations from the mean.

Range: 50 to 90

This includes almost all average, above-average, and below-average students.

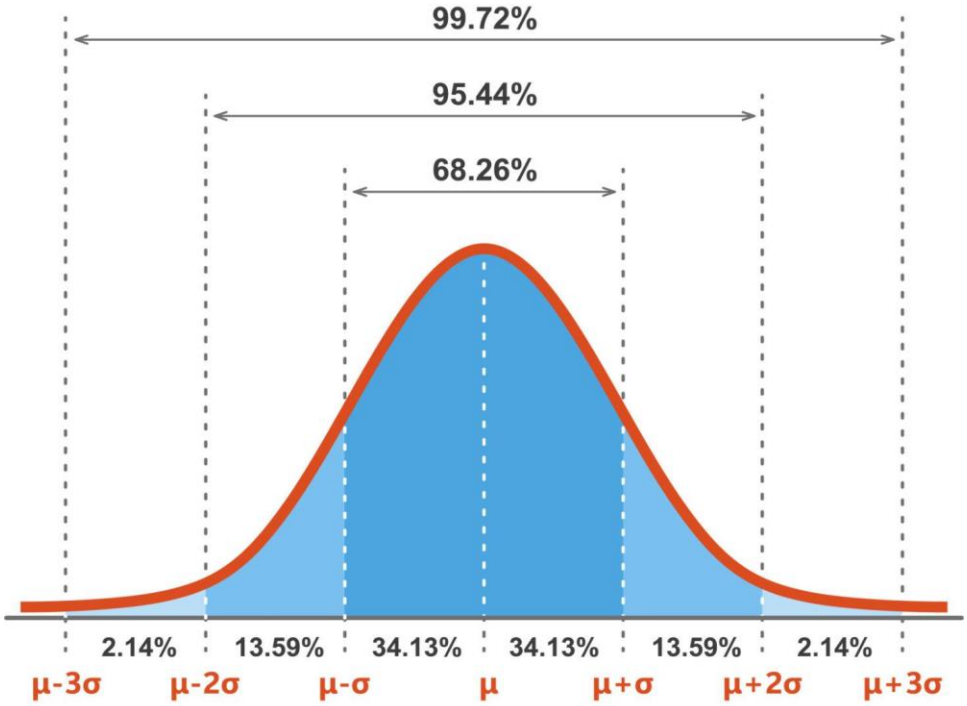
3. 99.7% Rule ($\mu \pm 3\sigma$)

Nearly **99.7% of students** score within three standard deviations.

Range: 40 to 100

Only very few students score below 40 or above 100 (extreme cases).

Graph Representation of Normal Distribution:



Components of the Graph

X-axis:
Represents student scores (low to high).

Y-axis:
Represents frequency (number of students).

Center (Mean μ):
Average student score.

Standard Deviation Points:

- $\mu - 1\sigma$ and $\mu + 1\sigma$
- $\mu - 2\sigma$ and $\mu + 2\sigma$
- $\mu - 3\sigma$ and $\mu + 3\sigma$

The curve is symmetric around the mean.

Interpretation of Student Performance

Using the empirical rule, student performance can be classified as:

Below -2σ

- Very low scores
- Few students
- Need improvement support

Between -1σ and $+1\sigma$

- Majority of students
- Average performance

Above $+1\sigma$

- High-performing students
- Above-average performance

Beyond $\pm 3\sigma$

- Extremely rare cases
- Exceptional or very weak performance

This helps teachers understand overall class performance.

Applications of Normal Distribution in Education

Normal distribution is useful in:

Performance analysis – Understanding overall student results
Grading on a curve – Assigning grades based on relative performance
Exam evaluation – Checking if the paper was too easy or difficult
Predicting results – Estimating expected outcomes
Identifying outliers – Detecting exceptionally high or low scorers

Advantages

- Easy to interpret
- Useful for prediction
- Represents real-world data effectively
- Helps in decision-making

Limitations

- Not all data follows normal distribution

- Sensitive to extreme values
- Requires sufficient data for accuracy

Conclusion

The normal distribution is a powerful statistical tool used to represent real-world data effectively. In the domain of student test scores, most students score near the average, forming a bell-shaped curve. The empirical rule explains that 68%, 95%, and 99.7% of values lie within one, two, and three standard deviations from the mean respectively.

By understanding the normal distribution and empirical rule, educators can analyze performance patterns, design fair grading systems, and make informed decisions. Therefore, the normal distribution plays a crucial role in educational data analysis and statistical interpretation.